

Case 07 / commit 5aa2184e: net/mlx5 HWS definer-layout double free

Exact predicate: report only when the conversion-failure edge still reaches the cleanup label that releases mt->fc before ownership is established.

```
M1 mlx5hws_definer_calc_layout(ctx, mt, match_definer) {
M2     struct mlx5hws_hl *match_hl;
M3     match_hl = kzalloc(sizeof(*match_hl), GFP_KERNEL);
M4     if (!match_hl)
M5         return -ENOMEM;
M6     ret = hws_definer_conv_match_params_to_hl(ctx, mt, match_hl);
M7     if (ret) {
M8         mlx5hws_err(ctx, "Failed to convert items to header layout");
M9         goto free_fc;
M10+    goto free_match_hl;
M11 }
M12 /* Find the match definer layout for header layout match union */
M13 ret = hws_definer_find_best_match_fit(ctx, match_definer, match_hl);
M14 if (ret)
M15     goto free_fc;
M16 free_fc:
M17     kfree(mt->fc);
M18+ free_match_hl:
M19     kfree(match_hl);
M20     return ret;
M21 }
```

1 owns mt->fc on success

2 failure -> free_fc

3 releases mt->fc

4 patch barrier

→ Solid = patch control flow
--- Dashed = checker-to-code mapping

Unsafe/report
Refined/barrier
Generic scan

#A. Patch hunk in
mlx5hws_definer_calc_layout

#B. Before refinement
Broad allocation/free checker

```
B1 Checker<PostCall, PreCall, Bind>
B2 AllocatedRegionMap<MemRegion, bool>
B3 checkPostCall(Call):
B4     if (!ExprHasName(Call, "kzalloc"))
B5         return;
B6     MR = getBaseRegion(Call.getOriginExpr());
B7     State = set<AllocatedRegionMap>(MR, true);
B8 checkPreCall(Call):
B9     if (!ExprHasName(Call, "kfree"))
B10        return;
B11     MR = Call.getArgSVal(0).getAsRegion();
B12     IsAllocated = State->get<AllocatedRegionMap>(MR);
B13     if (!IsAllocated || !*IsAllocated)
B14         report();
```

5 broad kfree rule

Strict validation
B:10/9; fixed noisy
C:1/0; fixed silent

#C. After refinement
Exact path/state predicate

```
C1 checkASTCodeBody(F):
C2     if F.name != "mlx5hws_definer_calc_layout"
C3         return;
C4     for each IfStmt S with condition "ret":
C5         Prev = previousStmt(S);
C6         require Prev is conversion step that may establish mt->fc
C7         require failure of Prev means mt->fc is not owned yet
C8         // anchor: hws_definer_conv_match_params_to_hl(...)
C9         Then = thenBranch(S);
C10        Target = cleanupTarget(Then);
C11        require Target releases mt->fc before match_hl cleanup
C12        // vulnerable anchor: free_fc -> kfree(mt->fc)
C13        require Target is not the patch barrier cleanup
C14        // fixed barrier: free_match_hl -> kfree(match_hl)
C15        report();
```

6 path/state predicate